



KATZ

Alumina Toughened Zirconia (ATZ)

KEY PROPERTIES

- Ultra-high strength
- Excellent toughness
- High surface finish ($R_a < 0.1 \mu\text{m}$)
- No aging (low temperature degradation)
- High wear resistance

APPLICATIONS

- Cutting tools
- Wear resistant tools for stamping
- Cold drawing dies
- Thread and wire guides
- High pressure components (plungers, casings)

MAIN PROPERTIES

Property	Value	Unit
Density	5.50	g/cm^3
Flexural Strength	1400	MPa
Fracture toughness	6.5	$\text{MPa}\cdot\sqrt{\text{m}}$
Young Modulus	260	GPa

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		1200 °C
Recomended temp		<1100 °C
Continuous temp		1000 °C

ELECTRICAL BEHAVIOR

Insulator

MATERIAL INFORMATION

Material type		ATZ (Alumina Toughened Zirconia)
Structure		Composite Ceramic
Machinability		Medium
Key feature		Ultra high strength
Color		White

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

Nanoker Research S.L.

Polígono de Olloniego, Parcela 21A, Nave 4-8, 33660, Oviedo (Asturias), Spain

(+34) 985 20 76 13 | info@nanoker.com